

CIAN ROCHE

 [cianmroche.github.io](https://github.com/cianmroche) |  roche@mit.edu |  0000-0002-3400-6991

EDUCATION

Massachusetts Institute of Technology <i>Ph.D. Candidate, Physics</i> Thesis: “ <i>The Dark Universe Through The Strong Lens: Cluster and Galaxy-Scale Probes of the Dark Sector</i> ” Advisors: Prof. Michael McDonald, Prof. Mark Vogelsberger	2021 – Present
Universität Tübingen <i>M.Sc., Mathematical Physics, 1.1 GPA[†]</i> Thesis: “ <i>Exact parallel waves in general relativity</i> ” Advisor: Prof. Carla Cederbaum	2019 – 2021
University College Cork <i>B.Sc., Physics, 1st class honours</i>	2015 – 2019

HONORS AND AWARDS

MIT Physics Henry W Kendall Teaching Fellowship	2026
DAAD Graduate Program Full Scholarship	2019
University College Cork College Scholar	2019
Max Planck Institute for Plasma Physics Research Scholarship	2018
Naughton Research Fellowship (one of 8 in Ireland)	2017
Naughton Foundation Scholarship (one of 29 in Ireland)	2015
All-Ireland Scholarship (one of 125 in Ireland; declined)	2015

TEACHING

Massachusetts Institute of Technology <i>Average student rating 6.95/7 — highest-rated[‡] across 55 semesters of recorded 8.13/8.14 evaluations.</i>	
Teaching Assistant, Experimental Physics II (8.14)	Spring 2026
Teaching Assistant, Experimental Physics I (8.13)	Fall 2024
Teaching Assistant, Experimental Physics I (8.13)	Spring 2024
Teaching Assistant, Experimental Physics I (8.13)	Fall 2023
Teaching Assistant, Experimental Physics I (8.13)	Fall 2022
Instructor - Gaia DR3 Hackathon	June 2022
University College Cork	
Teaching Assistant, Introductory Physics II	Spring 2018
Leader, Peer Assisted Learning Scheme	2016 – 2018

PUBLICATIONS

ADS library: <https://tinyurl.com/CianRocheADS> 45 first-author citations, 127 citations, *h*-index 7

First Author or Significant Contribution

7. **Roche, C.**, Sharon, K., McDonald, M., Vogelsberger, M. 2026; “*Mass Bias, Miscentering, and Magnification Uncertainties in Cluster Strong Lensing*”, (in prep)
8. Dutra, I., Natarajan, P., **Roche, C.** 2026; “*Where Cluster Galaxies Reside and How Strongly They Lens*”, (in prep)
7. **Roche, C.**, McDonald, M., Natarajan, P., Sharon, K., Dutra, I., Meneghetti, M., Vogelsberger, M. 2026; “*Host-Dependence and Line Of Sight Effects on Galaxy-Galaxy Strong Lensing in Clusters*”, submitted, *ApJ*, [arxiv:2608.30079](https://arxiv.org/abs/2608.30079)

[†]on a scale of 1-5, where 1 is the highest grade.

[‡]Among teaching staff with a minimum of 3 evaluations.

6. Fosdick, K., McDonald, M., Khullar, G., Gladders, M., Häußler, B., **Roche, C.**, et al. 2026; “*Quantifying the Emergence of Brightest Cluster Galaxies In SPT and Planck Clusters: A Novel Application of the Gini Index*”, accepted, *ApJ*
5. **Roche, C.**, Vogelsberger, M., McDonald, M., Dutra, I., Natarajan, P., Shen, X., Shreeram, X., Metcalf, R., Sharon, K., Birrer, S., Lee, W., Meneghetti, M. 2026; “*A Consistent Implementation of Cluster Strong Lensing in Cosmological Simulation Light Cones*”, accepted at OJA, [arXiv:2605.30433](#)
4. Tran, V., Shen, X., Vogelsberger, M., Gilman, D., O’Niell, S., **Roche, C.**, Zier, O., Gao, J. 2024; “*Novel Density Profile for Isothermal Cores of Dark Matter Halos*”, *Phys. Rev. D* 113, 063024, [arXiv:2411.11945](#)
3. **Roche, C.**, McDonald M., Vogelsberger, M., Shen, X., Springel, V., Hernquist, L., Pakmor, R., Bose, S., Kannan, R. 2024; “*Brightest Cluster Galaxy Offsets in Cold Dark Matter*”, OJA, astro-ph.GA 7, [arXiv:2402.00928](#)
2. **Roche, C.**, Necib, L., Lin, T., Ou, X., Nguyen, T. 2024; “*The Escape Velocity Profile of the Milky Way from Gaia DR3*”, *ApJ*, 972, 70, [arXiv:2402.00108](#)
1. **Roche, C.**, Aazami, A.B. & Cederbaum, C. 2023; “*Exact parallel waves in general relativity*” *Gen Relativ Gravit* 55, 40, [doi:10.1007/s10714-023-03083-x](#)

Nth Author

4. Lee, W., et al. (incl. **Roche, C.**) 2026; “*Evolution of a low-mass cluster merger and its radio relics in the self-interacting dark matter paradigm*”, (in prep)
3. Rose J. C., et al. (incl. **Roche, C.**) 2025; “*The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties*”, [arXiv:2512.00148](#)
2. Nguyen, T., et al. (incl. **Roche, C.**) 2024; “*How DREAMS are made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds*”, *ApJ*, 997, 336, [arXiv:2409.02980](#)
1. Rose J. C., et al. (incl. **Roche, C.**) 2024; “*Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations*”, *ApJ*, 982, 68, [arXiv:2405.00766](#)

Non-Refereed

1. **Roche, C.** & Garnavich, P. 2020; “*Testing Progenitor Models Using the Late-Time Light Curve of Supernova 1992A*” *Res. Notes AAS* 4 207, [doi:10.3847/2515-5172/abcb05](#)

PRESENTATIONS

Invited (Colloquia/Seminars)

Talk	“ <i>Are Cluster Galaxies Too Good at Lensing?</i> ” Harvard University	Dec 2026
Talk	“ <i>Illuminating the Dark Sector with Strong Gravitational Lensing</i> ” Stanford University	Oct 2026
Talk	“ <i>Are Cluster Galaxies Too Good at Lensing?</i> ” MIT Kavli Institute for Astrophysics and Space Research	Oct 2026
Talk	“ <i>Illuminating the Dark Sector with Strong Gravitational Lensing</i> ” Stony Brook University	Sep 2026
Talk	“ <i>Illuminating the Dark Sector with Strong Gravitational Lensing</i> ” University of Michigan	Sep 2026
Talk	“ <i>Light Cone Strong Lensing in Cosmological Hydrodynamical Simulations</i> ” MIT Kavli Institute for Astrophysics and Space Research	Mar 2026
Talk	“ <i>Understanding Dark Matter with Simulated Strong Lensing</i> ” Yale University	Sep 2025

Talk	“ <i>Dark Matter Self-Interaction via Galaxy Cluster Dynamics</i> ” Harvard University	Feb 2024
Talk	“ <i>Galaxies Lacking Dark Matter Produced in a Cosmological Simulation</i> ” MIT Kavli Institute for Astrophysics and Space Research	Apr 2022
Lecture	“ <i>Waves in General Relativity</i> ” Mathematical Relativity Class – University of Tübingen	July 2021
Talk	“ <i>Exact parallel waves in general relativity</i> ” Mathematical Physics Colloquium – University of Tübingen	June 2021
Talk	“ <i>How to Stop a Hurricane: Complexity in Physics</i> ” “Blow Your Mind Week” 2020 – University College Cork	Oct 2020
Talk	“ <i>Hermiticity of Young Operators</i> ” Young Tableaux Seminar Series – Universität Tübingen	June 2020

Contributed

Poster	“ <i>Are Cluster Galaxies Too Good at Lensing?</i> ” Probe Combination for Dark Matter Physics in the Era of Large Surveys – Kavli Institute for Cosmological Physics, University of Chicago	Sep 2026
Poster	“ <i>The Line of Sight as a Major Systematic of Cluster Strong Lensing</i> ” Gravitational Lensing at the Turning Point – Sexten Center for Astrophysics	July 2026
Talk	“ <i>The Observation-Simulation Connection and the Line of Sight in Cluster Strong Lensing</i> ” Strong Gravitational Lensing Workshop – University of Montpellier	June 2026
Talk	“ <i>Consistent Strong Lensing Predictions from Cosmological Simulations</i> ” Scaling up Lensing Workshop – University of Liège	Oct 2025
Poster	“ <i>Unveiling Dark Matter: Testing CDM with the First End-to-End Cluster Strong Lensing Simulations</i> ” Tracing cosmic Evolution with Galaxy Clusters V – Sexten Center for Astrophysics	July 2025
Talk	“ <i>Testing CDM with the First End-to-End Cluster Strong Lensing Simulations</i> ” European Astronomical Society Meeting 2025 – University College Cork	June 2025

ADVOCACY AND OUTREACH

MIT GAGA Initiative Co-chair 2022 – 2023 | Organizing Committee 2021 – 2024
The Graduates Advising Graduate Admissions program offers information and mentorship to underrepresented groups in physics, and advises on improvements to the admissions process.

MIT PhysGAAP Program 2021 – 2023
The Physics Graduate Application Assistance Program provides guidance to prospective applicants, and liaises with physics department leadership on issues of accessibility in the graduate application. I have acted as one of the primary organizers and have mentored 5 students through the program.

Astrogazers, MIT 2024 – Present
Student Academic Representative, University College Cork 2017 – 2019

LEADERSHIP AND SERVICE

Admissions Committee Member, MIT Summer Research Program 2023 Cycle
The MIT Summer Research Program aims to provide high-quality research experiences to promising undergraduate students from underrepresented backgrounds.

Admissions Committee Member , MIT Physics Graduate Program	2023 Cycle, 2024 Cycle
Volunteer , AAS 241st Meeting Graduate School Fair	2023
Graduate Student Office Manager , MIT Kavli Institute	2022 – 2025
Organizer , MIT Kavli Institute Graduate Student Lunch Series	2022 – 2023
Member , MIT Physics Graduate Student Council	2021 – 2023
Organizing Committee , MIT Graduate Student Union	2021 – 2023
Astronomy Officer , UCC Physics and Astronomy Society	2017 – 2019

EDUCATIONAL RESOURCES

These are some of the educational resources and tools I have produced to support my peers, my students or the community at large.

figinvert

A lightweight command line tool based in python that inverts vector and raster graphics for dark mode talks. In the default mode, mimics overleaf’s PDF inversion.

Caustica

An educational and intuition-building online strong lensing simulator, also capable of producing high-quality lensing media such as video and vector graphics.

χ by Eye

A learning tool in a geoguessr-like format for training students to recognize believable or unrealistic statistics in a plot by eye. Sandbox mode is particularly useful for gaining hands-on intuition.

AstroWiki

I created and now co-maintain a set of living notes covering most of graduate level astrophysics, specifically designed to aid the next generations of graduate students when studying for the MIT astrophysics oral qualification exam.